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Artikel Ilmiah 1



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Journal	Technologies
Special Issue	Advanced Technologies of Biomedical Image Processing
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Keywords	Child Malnutrition Detection;Machine Learning;Support Vector Machine;Particle Swarm Optimizer
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Abstract 10

Stunting remains a persistent public health challenge in low- and middle-income countries, where growth-monitoring at the primary healthcare level still relies heavily on manual anthropometric assessment prone to human error and inconsistent interpretation. Machine learning (ML) approaches proposed in prior work often demand computational resources unavailable in such settings, limiting their practical adoption for field-level screening. This study proposes SVM-PSO, a hybrid framework that couples Support Vector Machines with Particle Swarm Optimization to automatically tune the regularization parameter C and kernel coefficient gamma, reducing the manual trial-and-error that typically constrains SVM performance. The framework is evaluated on two independent datasets: a structured anthropometric dataset of 6,500 child records and an image dataset of 1,859 samples, the latter processed through frozen VGG16 feature extraction prior to classification. Model performance is assessed using repeated stratified k-fold cross-validation paired with Wilcoxon signed-rank testing, rather than a single train-test split, to support statistically defensible comparisons. On the tabular dataset, SVM-PSO reached 96.81% accuracy and 98.05% AUC, matching a random-search baseline (SVM-RO) within statistical uncertainty while requiring substantially less computation. On the image dataset, SVM-PSO reached 85.98% accuracy and 89.79% AUC, significantly outperforming both the default-parameter SVM and the random-search baseline. Feature-level interpretability (SHAP) and calibration analysis were conducted for the tabular model, and both trained configurations were deployed as a preliminary web-based interface to illustrate a potential pathway toward field-level stunting screening, though further validation with front-line health workers remains necessary.

Keywords: Child Malnutrition Detection; Machine Learning; Support Vector Machine; Particle Swarm Optimizer

1. Introduction 36

Stunting remains one of the most persistent global health challenges, particularly in low- and middle-income countries, where it often originates from chronic malnutrition during the first 1,000 days of life [1]. In Indonesia specifically, national stunting prevalence has remained high over the past decade despite sustained government intervention programs, with consistent evidence pointing to nonexclusive breastfeeding, low

Require: Coefficients $w, \mathcal{C}_1, \mathcal{C}_2$ 226

Require: Search space bounds for $(\mathcal{C}, \gamma)$ 227

Ensure: Optimal hyperparameters $(\mathcal{C}^*, \gamma^*)$, final model $f_{svm-pso}$ 228

- 1: Initialize population $\mathcal{P} = \{x_1, \dots, x_N\}$ with random positions within bounds 229
- 2: Initialize velocities $\mathcal{V} = \{v_1, \dots, v_N\}$ randomly 230
- 3: Initialize personal best $p_i \leftarrow x_i$ for all i , and global best $g \leftarrow \operatorname{argmax} \operatorname{fitness}(\mathcal{P})$ 231
- 4: for $i = 1$ to T do 232
- 5: for $i = 1$ to N do 233
- 6: Evaluate $\operatorname{fitness}(x_i)$ via mean k-fold cross-validation accuracy 234
- 7: Train SVM(x_i) on $\mathcal{D}_{train}$ using k-fold CV 235
- 8: Compute mean classification accuracy Acc from validation folds 236
- 9: if $\operatorname{fitness}(x_i) > \operatorname{fitness}(p_i)$ then 237
- 10: $p_i \leftarrow x_i$ 238
- 11: if $\operatorname{fitness}(x_i) > \operatorname{fitness}(g)$ then 239
- 12: $g \leftarrow x_i$ 240
- 13: for $i = 1$ to N do 241
- 14: Draw $r_1, r_2 \sim \mathcal{U}(0, 1)$ 242
- 15: $v_i \leftarrow w \cdot v_i + x_i \cdot r_1 \cdot (p_i - x_i) + \mathcal{C}_1, \mathcal{C}_2 \cdot (g - x_i)$ 243
- 16: $x_i \leftarrow x_i + v_i$ 244
- 17: $x_i \leftarrow \operatorname{ApplyBoundaryHandling}(x_i)$ 245
- 18: Construct final SVM model f_{svm-p} using best hyperparameters $g = (\mathcal{C}^*, \gamma^*)$ 246
- 19: return f_{svm-p}, g 247

2.4. Metric Evaluation 248

The proposed framework is assessed using five standard classification metrics: Accuracy, Recall, Precision, F1-Score, and Area Under the ROC Curve (AUC). Pairwise differences between models were tested for statistical significance using the Wilcoxon signed-rank test on fold-level scores, rather than relying on point estimates from a single split. 249 250 251 252 253

2.5. Experiment Setting 254

Model evaluation was conducted using stratified 5-fold cross-validation repeated three times with different random partitioning. This is not a "15-fold" scheme in the conventional sense; rather, it is a standard 5-fold split re-executed under three independent random shuffles, which produces 15 paired accuracy observations per model and enables formal statistical significance testing (Section 3.4) that a single 5-fold or 10-fold run cannot support reliably. A single fold split, however common in prior work on similar tasks, leaves too few data points for a Wilcoxon signed-rank test to be robust against the influence of one unusually favorable or unfavorable partition. The Tabular Dataset were each evaluated under this protocol independently, given that the two modalities differ substantially in sample size, feature dimensionality, and label balance. 255 256 257 258 259 260 261 262 263 264

Experiments were implemented in Python using scikit-learn 1.3.0 for the SVM classifier and cross-validation pipeline, TensorFlow 2.13.0 for VGG16 feature extraction, and SciPy for the Wilcoxon signed-rank tests. Image preprocessing was handled with OpenCV 4.8.0, and the deployed web application was built with Gradio, hosted on Hugging Face Spaces. 265 266 267 268 269

Artikel Ilmiah 2



#67 (1571345184): Interpretable Machine Learning for Child Stunting Assesment

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Authors	Drag to change order	Author name	Author affiliation (edit for paper)	Author email	Email	Delete
	⋮	Haerul Yuda Aditiya	Universitas Buana Perjuangan Karawang, Indonesia	if23.haeruladitiya@mhs.ubpkarawang.ac.id	✉	✕
	⋮	Tatang Rohana	Universitas Buana Perjuangan Karawang, Indonesia	tatang.rohana@ubpkarawang.ac.id	✉	✕
	⋮	Sinta Maria Dewi	Universitas Buana Perjuangan Karawang, Indonesia	sintamaria@ubpkarawang.ac.id	✉	✕
	⋮	Ahmad Khusaeri	Universitas Singaperbangsa Karawang, Indonesia	ahmad.khusaeri@cs.unsika.ac.id	✉	✕
	⋮	Oman Komarudin	Universitas Singaperbangsa Karawang, Indonesia	oman@unsika.ac.id	✉	✕
	⋮	Syntia Widyayuningtias Putri Listio	Universitas Bhinneka Nusantara, Indonesia	syntia@stiki.ac.id	✉	✕
	⋮	Candra Zonyfar	Sun Moon University, Korea (South) & Universitas Buana Perjuangan Karawang, Indonesia	candra@ubpkarawang.ac.id	✉	✕



Full paper title

Interpretable Machine Learning for Child Stunting Assesment [🔗](#)

Conference and track

[The 2026 International Conference on Data Engineering and Enterprise System](#) - [The 2026 International Conference on Data Engineering and Enterprise System](#)

Abstract

 [🔗](#) Stunting among children under five remains a major public health challenge in Indonesia, demanding...

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Interpretable Machine Learning for Child Stunting Assessment

1st Haerul Yuda Aditiya
Faculty of Computer Science
Universitas Buana Perjuangan
Karawang
Karawang, Indonesia

4th Oman Komarudin
Faculty of Computer Science
Universitas Singaperbangsa Karawang
Karawang, Indonesia

6th Tatang Rohana
Faculty of Computer Science
Universitas Buana Perjuangan
Karawang, Indonesia

2nd Sinta Maria Dewi
Faculty of Education
Universitas Buana Perjuangan
Karawang
Karawang, Indonesia

5th Syntia Widyayuningtias Putri Listio
Faculty of Science and Technology
Universitas Bhinneka Nusantara
Malang, Indonesia

3rd Ahmad Khusaeri
Faculty of Computer Science
Universitas Singaperbangsa Karawang
Karawang, Indonesia

7th Candra Zonyfar
Department of computer science and engineering
Sun Moon University
Asan , South Korea
Faculty of Computer Science
Universitas Buana Perjuangan
Karawang, Indonesia

Abstract—Stunting among children under five remains a major public health challenge in Indonesia, demanding solutions that support both province-level surveillance and individual-level nutritional screening to guide targeted intervention. However, existing approaches typically address these needs in isolation, relying on isolated, single-purpose models that lack interpretability and fail to concurrently address regional nutritional vulnerability alongside individual anthropometric status. Here, we propose an interpretable, integrated machine learning framework that combines clustering, regression, and classification as complementary but independently trained components to comprehensively assess child stunting across spatial and individual scales. The framework draws on two distinct datasets: a provincial spatial-demographic dataset (34 provinces and 514 districts/cities) and an individual child anthropometric dataset, capturing regional nutritional indicators and individual growth measures, respectively. In the clustering task, we evaluated K-Means, Gaussian Mixture Models, and Hierarchical clustering using Elbow and Silhouette analysis, and identified Hierarchical clustering as the best-performing method, achieving a Silhouette score of 0.3199 and a Davies-Bouldin Index of 0.9877. Six regression algorithms were benchmarked to predict Height-for-Age Z-scores (HAZ) and Weight-for-Age Z-scores (WAZ), among which Random Forest and CatBoost achieved good performance. Nutritional status classification compared five

I. INTRODUCTION

Stunting among children under five remains a significant global health challenge, with 96.7% of the approximately 149 million cases recorded in 2020 occurring in low and middle income countries [1], [2]. In Indonesia, the 2024 Indonesian Nutritional Status Survey (SSGI) [3] reported a national stunting prevalence of 19.8%, close to the 20% threshold defined by the World Health Organization as indicative of a public health problem, while several provinces in eastern Indonesia still recorded prevalence rates exceeding 30%, a pattern already evident in the 2018 Basic Health Research (Riskesdas) data showing a national figure of 30.8% [4]. This persistent burden stands in contrast to the target set by the National Medium Term Development Plan (RPJMN) [5], which mandates a reduction of national stunting prevalence to below 14% by 2029. As an archipelagic country with substantial geographic, social, economic, and health related heterogeneity across regions, Indonesia faces the additional challenge that uniform, one size fits all intervention strategies often fail to address region specific determinants of child malnutrition, underscoring the need to understand both the spatial distribution and temporal trends of stunting risk across administrative areas.

case.

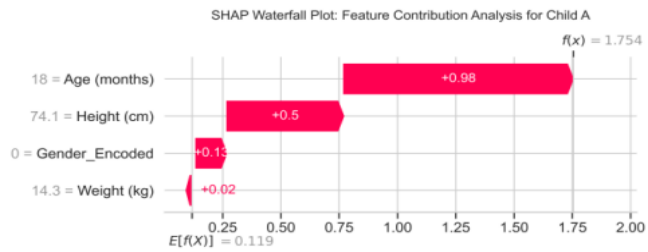


Fig. 8. SHAP waterfall plot illustrating individual feature contribution for a representative child case.

For this individual case, with an age of 18 months, a height of 74.1 cm, Gender_Encoded of 0, and a weight of 14.3 kg, Age contributed the largest positive impact on the predicted output (+0.98), followed by Height (+0.5), Gender_Encoded (+0.13), and Weight (+0.02). The cumulative contribution of these four features shifted the prediction from the baseline expected value of 0.119 to a final output of 1.754, illustrating how individual feature values combine to drive a single prediction, consistent with the global feature importance pattern observed in Figure 7, where Age and Height are the dominant contributors.

D. Dashboard Implementation

To translate the analytical results into a practical decision support tool, the trained clustering, regression, and classification models were deployed within an interactive web based dashboard (see Figure 9), consisting of three main interfaces as described in Section II.

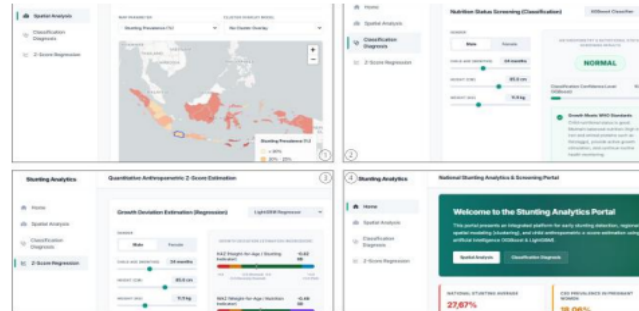


Fig. 9. Interface of the web dashboard. (1) Clusterin; (2) classification; (3) regression; (4) Home page.

IV. CONCLUSION

This study proposed an interpretable, integrated machine learning framework for child stunting assessment, combining clustering, regression, and classification within a single unified pipeline supported by two distinct datasets, the

and classification (99.98% accuracy) tasks, with CatBoost a close second throughout. What ties these results together is less the specific algorithm choice than the recurring dominance of height and age as predictive features, a signal reinforced independently by SHAP analysis on both the LightGBM regressor and XGBoost classifier, suggesting the underlying growth-status signal is robust across model families rather than an artifact of any single algorithm.

Several limitations should be acknowledged. The clustering task relies on a relatively small number of observations at the provincial level, which may explain the non-monotonic behavior observed in the Within-Cluster Sum of Squares curve and limits the extent to which the resulting risk clusters can be generalized beyond the observed period. The CAdataset, while large in size, is sourced from a single public dataset rather than a nationally representative individual level survey, which may limit the generalizability of the regression and classification results to the broader Indonesian child population. In addition, this study does not establish a direct quantitative link between the regional level clustering results and the individual level regression and classification results, as the two tasks rely on separate datasets.

Future work may address these limitations by incorporating a nationally representative individual level dataset, extending the temporal coverage of the regional level data to examine trends over time, and exploring methods to more directly connect regional vulnerability profiles with individual level nutritional outcomes within a single integrated model.

DATA AND CODE AVAILABILITY

The source code for the experiments conducted in this study is publicly available in a GitHub repository. The resulting interactive dashboard is also publicly accessible. Source code: github.com/haerulyudaaditiya/Stunting-Detection-ML-Spatial-Analysis. Dashboard: haerulyudaaditiya.github.io/Stunting-Detection-ML-Spatial-Analysis

ACKNOWLEDGMENT

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- [1] UNICEF/WHO/WORLD BANK, "Levels and trends in child malnutrition UNICEF / WHO / World Bank Group Joint Child Malnutrition Estimates Key findings of the 2021 edition," *World Heal. Organ.*, pp. 1–32, 2021, [Online]. Available: <https://www.who.int/publications/i/item/9789240025257>

Artikel Ilmiah 3



Machine Learning in Child Stunting Detection: A Comprehensive Survey of Anthropometric and Image-Based Approaches

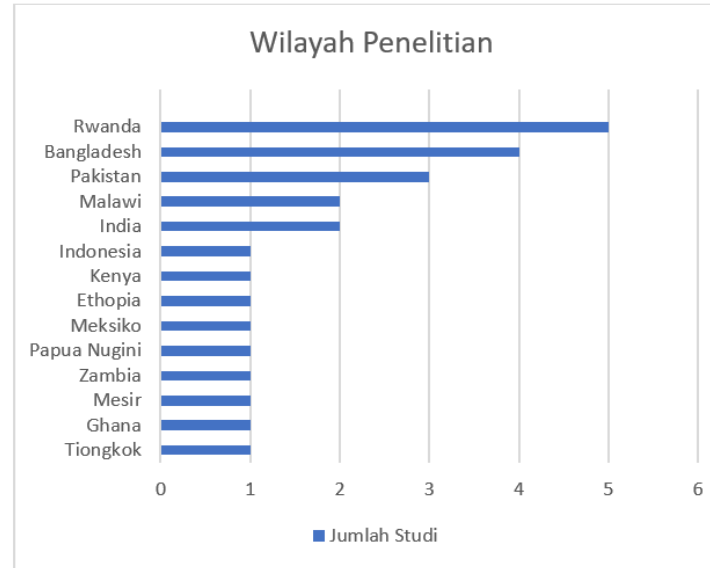
Abstract

Stunting pada anak usia dibawah lima tahun masih menjadi tantangan kesehatan masyarakat utama di berbagai negara, khususnya negara berpendapatan rendah dan menengah, sehingga deteksi dini menjadi kunci untuk mendukung intervensi gizi yang tepat waktu. Prediksi risiko stunting berbasis Artificial Intelligence/Machine Learning (AI/ML) dapat dimanfaatkan oleh peneliti maupun tenaga kesehatan untuk mengidentifikasi anak berisiko sejak dini berdasarkan karakteristik anak, orang tua, rumah tangga, hingga lingkungan. Oleh karena itu, topik ini semakin banyak diteliti oleh komunitas ilmiah, dengan berbagai pendekatan komputasional, sumber data, serta strategi evaluasi model yang beragam. Namun demikian, masih terbatas kajian sistematis yang merangkum dan mengorganisasi proposal-proposal yang telah ada pada topik ini. Akibatnya, peneliti yang ingin mengusulkan solusi AI/ML baru maupun praktisi kesehatan yang ingin menerapkannya berpotensi kesulitan memahami keragaman studi yang tersedia. Untuk mendukung profil tersebut, penelitian ini menyajikan Systematic Literature Review mengenai penerapan AI/ML untuk prediksi stunting. Kajian ini menyoroti meningkatnya komunitas ilmiah pada topik ini, di dominasi oleh algoritma ensemble seperti Random Forest dan XGBoost sebagai model performa terbaik, serta ketergantungan pada data survey kesehatan populasi seperti DHS. Meskipun demikian, penerapan dan pengujian model pada layanan kesehatan masih sangat jarang dilakukan, dan aspek generalisasi model cross-sectionanl populasi masih sering terabaikan.

1. Introduction

Stunting merupakan salah satu bentuk malnutrisi yang masih menjadi permasalahan kesehatan global. Kondisi ini ditantai dengan gangguan pertumbuhan pada anak dan banyak dikadi pada kelompok anak di bawah lima tahun karena berkaitan dengan berbagai karakteristik individu, ibu, rumah tangga, dan lingkungan. Penelitian yang dilakukan berbagai negara menunjukkan bahwa faktor-faktor yang berkaitan dengan stunting dapat berasal dari karakteristik anak, seperti usia, jenis kelamin, berat atau ukuran saat lahir. Selain itu, faktor ibu dan rumah tangga juga sangat berperan, seperti pendidikan ibu, status gizi ibu, kondisi ekonomi, urutan kelahiran, fasilitas sanitasi, dan sumber air (Islam et al., 2024; Mgonezulu et al., 2025; S. Ndagijimana et al., 2022). Kompleksitas faktor yang berkaitan dengan stunting membuat penelitian mengenai identifikasi dan prediksi stunting perlu mempertimbangkan berbagai aspek, tidak hanya karakteristik pertumbuhan anak. Oleh karena itu, sejumlah penelitian mulai memanfaatkan informasi dari aspek sosial, ekonomi, kesehatan, dan lingkungan yang tersedia dalam berbagai sumber data. Beberapa penelitian menggunakan data dari Demographic and Health Survey (DHS), Multiple Indicator Cluster Survey (MICS), *Living Standards Measurement Study–Integrated Surveys on Agriculture* (LSMS-ISA),

4.1.2 Distribusi negara/wilayah penelitian

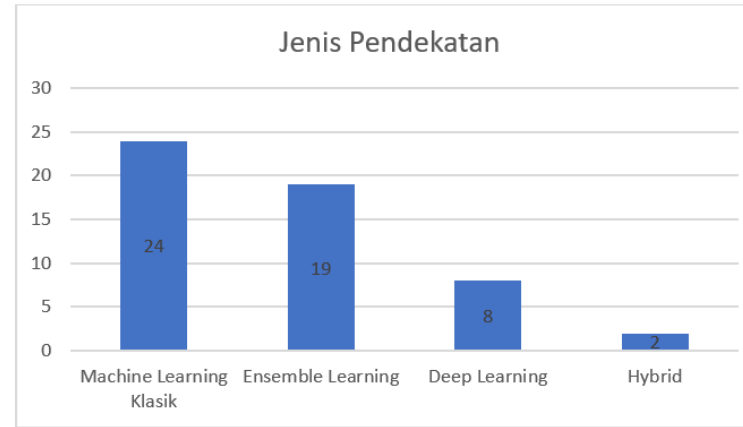


Gambar 2. Negara/wilayah penelitian berdasarkan jumlah studi

Studi terpilih berasal dari berbagai negara yang berpendapatan rendah dan menengah di Afrika, Asia, dan Kawasan Pasifik. Penelitian paling banyak dilakukan di Afrika Sub-Sahara, khususnya Rwanda yang ditemukan pada lima studi (Munyemana et al., 2024; A. Ndagijimana et al., 2025; S. Ndagijimana et al., 2022, 2025; Nduwayezu et al., 2025), diikuti Malawi (Mgomezulu et al., 2025; Mkungudza et al., 2024) dan Ethiopia (Asgedom et al., 2024). Negara Afrika lainnya meliputi Zambia (Chilyabanyama et al., 2022), Ghana (Anku & Duah, 2024), dan Kenya (Ogwel et al., 2024). Di Asia Selatan, terdiri dari Bangladesh (Haque et al., 2023; Islam et al., 2024; Rashid et al., 2026; Tamanna et al., 2025), Pakistan (Shahid et al., 2026; Usman & Kopczewska, 2022; Zhang et al., 2024), dan India (Rao et al., 2026; Sinharoy et al., 2026). Studi lainnya berasal dari Papua Nugini (Shen et al., 2023), Mesir (Hendy et al., 2025), Meksiko, (Fogarty et al., 2025) Tiongkok (Xiong et al., 2024), dan Indonesia (Widhari et al., 2024). Selain penelitian berbasis satu negara, terdapat juga studi lintas wilayah, seperti kawasan Afrika Timur (Jemil et al., 2026), 22 negara di Sub-Shara Afrika (Bachwenkizi et al., 2026), dan beberapa negeri

4.2 AI/ML approaches employed for stunting prediction

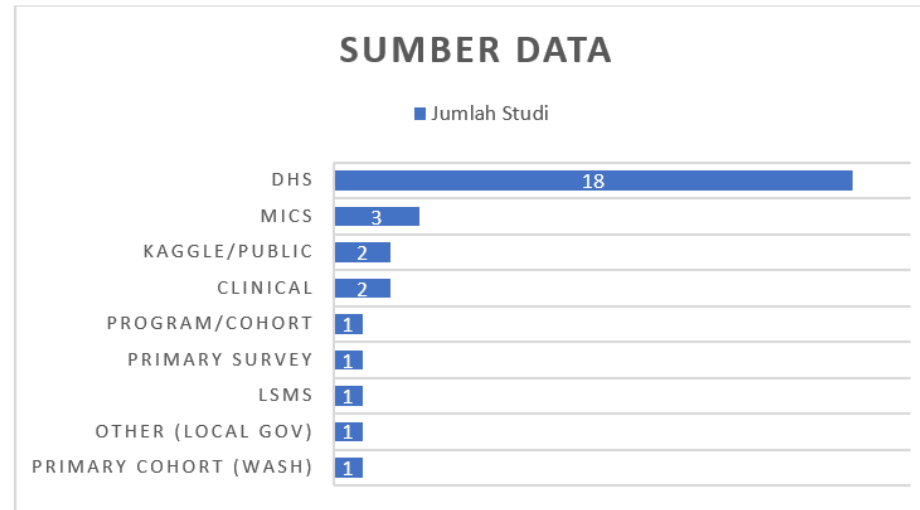
4.2.1 Jenis Pendekatan



Gambar 3. Jenis pendekatan berdasarkan jumlah studi

Berbagai jenis pendekatan digunakan dalam studi-studi terpilih. Pendekatan machine learning klasik, seperti regresi logistik, SVM, KNN, Naïve Bayes, Decision Tree/CART, dan Probit Regression merupakan pendekatan yang paling banyak digunakan, yaitu ditemukan pada 24 dari 30 studi (80%) (Chilyabanyama et al., 2022; Jemil et al., 2026; S. Ndagijimana et al., 2022; Widhari et al., 2024). Kedua, ensemble learning seperti Random Forest, XGBoost, dan Gradient Boosting digunakan pada 19 dari 30 studi kasus (63%) dan pada beberapa studi menjadi model dengan performa terbaik (Bachwenkizi et al., 2026; Fogarty et al., 2025; Munyemana et al., 2024). Ketiga, deep learning berupa Artificial Neural Network (ANN), Multilayer Perceptron (MLP), atau Deep Neural Network (DNN) digunakan pada 8 dari 30 studi (27%) (Fogarty et al., 2025; Islam et al., 2024; Mgomezulu et al., 2025; S. Ndagijimana et al., 2025; Ogwel et al., 2024; Sinharoy et al., 2026; Tamanna et al., 2025; Wudu et al., 2026). Keempat, pendekatan secara hybrid diterapkan pada 2 dari 30 studi (7%), yaitu kombinasi Extra Tree dengan Multilayer Perceptron (Wudu et al., 2026) dan Fire Hawk Optimizer K-Means dengan Enhanced Gradient Boosting Fuzzy berbasis XGBoost (Alam et al., 2025). Secara umum, machine learning klasik dan ensemble learning merupakan dua kategori yang paling sering digunakan dibandingkan deep learning maupun hybrid.

4.2.2 Algoritma/model spesifik yang paling sering digunakan



Gambar 5. Sumber data

Sumber data yang paling banyak digunakan oleh studi-studi terpilih yaitu survey kesehatan berskala nasional jenis Demographic and Health Survey (DHS) yang disesuaikan dengan wilayah penelitian, seperti RDHS, ZDHS, BDHS, EDHS, Mini-EDHS, MDHS, PNG DHS, Egypt DHS, dan NFHS-5, yang digunakan pada 18 studi (Chilyabanyama et al., 2022; S. Ndagijimana et al., 2022; Rao et al., 2026; Shen et al., 2023; Wudu et al., 2026). Multiple Indicator Cluster Survey (MICS) digunakan pada 3 studi (Anku & Duah, 2024; Usman & Kopczewska, 2022; Zhang et al., 2024). Dataset public seperti Kaggle digunakan pada 2 studi (Alam et al., 2025; Widhari et al., 2024). Begitu juga penggunaan data klinis/kohort yang digunakan pada 2 studi (Ogwel et al., 2024; Xiong et al., 2024). Sumber data yang lain masing-masing hanya ditemukan pada 1 studi seperti Living Standards Measurement Survey (LSMS) dari World Bank di Malawi (Mgomezulu et al., 2025), data program gizi Suchana di Bangladesh (Haque et al., 2023), survey primer cross-sectional di Rwanda (A. Ndagijimana et al., 2025), data pemerintah daerah di Aceh, Indonesia (Hasdyna et al., 2024), dan Kohort WASH di India (Sinharoy et al., 2026).